PROV LOGIC

50 MUST KNOW STATIC TIMING ANALYSIS INTERVIEW QUESTIONS

Practical, Theoretical & Tool-Level Questions
For VLSI Engineers to land in their Dreams Companies

intel

Qualcomm

SAMSUNG

AMD



Prasanthi Chanda

1.What is Setup Time Violation?

Company: Synopsys, Qualcomm

- **Setup time is the minimum duration for which data must remain stable before the active clock edge arrives at a flip-flop.**
- **A setup time violation happens when data reaches the flip-flop too late, i.e., after the required setup time window.**
- **This can result in incorrect data being captured, leading to functional errors.**
- **Setup violations typically occur due to high path delay in the combinational logic between two flip-flops.**
- **Timing tools check the maximum delay along a path to detect setup violations.**
- **Fixes include:**
  - **Reducing logic depth**
  - **Adding buffers to balance delay**
  - **Cell upsizing or restructuring the logic**

2. What is Clock Skew and how does it impact Setup and Hold Timing?

Company: Intel, Texas Instruments

- **Clock skew is the difference in clock arrival time at the launching and capturing flip-flops of a data path.**
- **If the clock arrives late at the destination, it's called positive skew; if it arrives early, it’s called negative skew.**
- **Positive skew can relax setup time requirements but may cause hold violations.**
- **Negative skew does the opposite: it tightens setup timing but can help with hold timing.**
- **Excessive skew can create both setup and hold issues on the same path.**

- **Skew arises from imbalanced clock routing, varying buffer delays, or inconsistent load capacitance.**
- **Fix strategies include:**
  - **Using Clock Tree Synthesis (CTS) to balance skew**
  - **Applying clock buffering**
  - **Leveraging useful skew to intentionally adjust timing paths**

3. A path is showing a Setup Violation. How will you debug and fix it?

Company: Apple, Broadcom

- **First, identify the violating path using STA reports (report_timing in PrimeTime).**
- **Check the startpoint and endpoint flip-flops, and trace the data path.**
- **Examine the combinational logic delay between the flip-flops.**
- **Analyze the clock arrival times to check for excessive clock skew or latency.**
- **Check for high fan-out nets, weak drive strengths, or long wire delays.**
- **Apply fixes such as:**
  - **Buffer insertion to reduce net delay**
  - **Cell upsizing for faster transitions**
  - **Logic restructuring to reduce path depth**

4. What is Hold Time Violation? How is it different from Setup Violation?

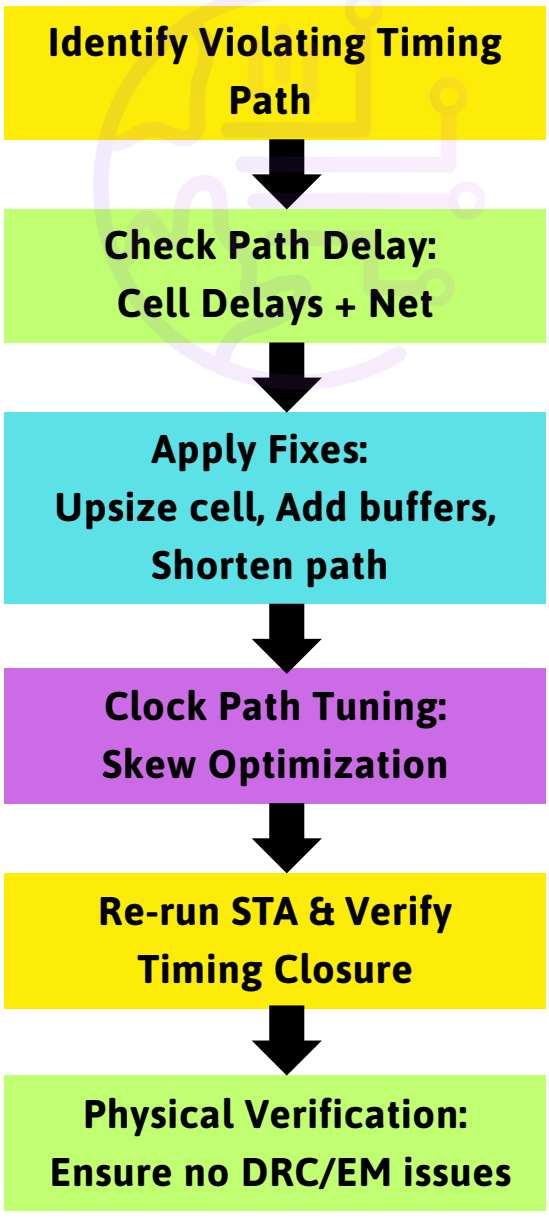
Company: Apple, Broadcom

- **Hold time is the minimum time data must be stable after the clock edge to be correctly latched.**
- **A hold violation occurs when data changes too soon after the clock edge, before the latch captures it.**

- This is a min delay violation, unlike setup which is a max delay violation.
- Hold violations are mostly due to fast paths, like:
  - Short logic paths
  - Strong driving cells
  - Low-capacitance nets
- Fixes include:
  - Inserting delay buffers in the data path
  - Cell downsizing to slow down logic transitions

5. How do you fix a Setup Violation?

Company: Nvidia, MediaTek



6.What is Static Timing Analysis (STA), and how is it different from Dynamic Timing Simulation?

Company: ARM, Texas Instruments

STA (Static Timing Analysis)

- **Analyzes timing of a digital circuit without applying vectors.**
- **Works on cell delays + interconnect delays using worst/best case conditions.**
- **Checks for setup, hold, recovery, and removal violations.**
- **Uses constraints like clock definitions, false paths, and multicycle paths.**

Dynamic Simulation

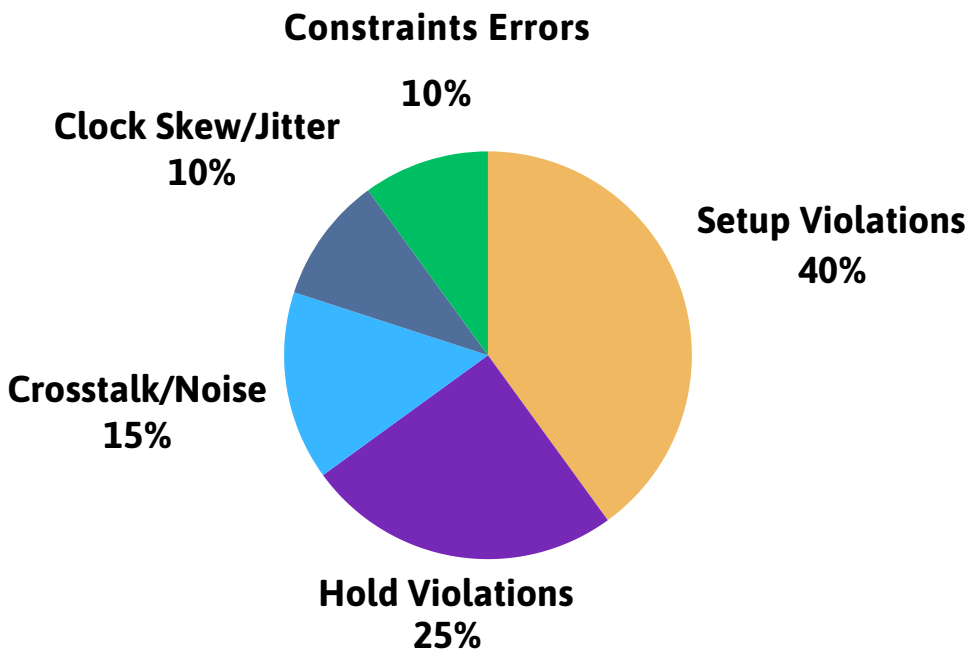
- **Uses real input vectors to simulate timing behavior.**
- **Captures functionality + timing using RTL or gate-level testbenches.**
- **Time-consuming and vector-dependent — may miss corner cases.**

Key Difference:

- **STA is fast, corner-based and exhaustive for timing; dynamic sim checks functionality but is limited by vector coverage.**

7. What percentage of STA issues arise from each category in a real design?

Company: Qualcomm, Synopsys



8.What is the Importance of Matching in Analog Layout Design?

Company: Marvell, Texas Instruments

Parameter	Setup Violation	Hold Violation
Definition	Data arrives after the setup window	Data changes too soon after clock
Type of Check	Maximum delay check	Minimum delay check
Occurs When	Data path is too slow	Data path is too fast
Common Fixes	Buffer insertion, path restructuring	Delay buffers, downsizing cells
Clock Consideration	Setup uses propagated clock	Hold uses zero-delay clock
STA Focus	Late data arrival	Early data arrival
Impact	Can cause incorrect latch data	Can overwrite previous data

9. What are the main components of a timing path in STA?

Company: Apple, Qualcomm

Components of a Timing Path:

1.Launch Clock Edge

- **The clock edge that triggers data launch from the source flip-flop.**

2.Data Path Delay

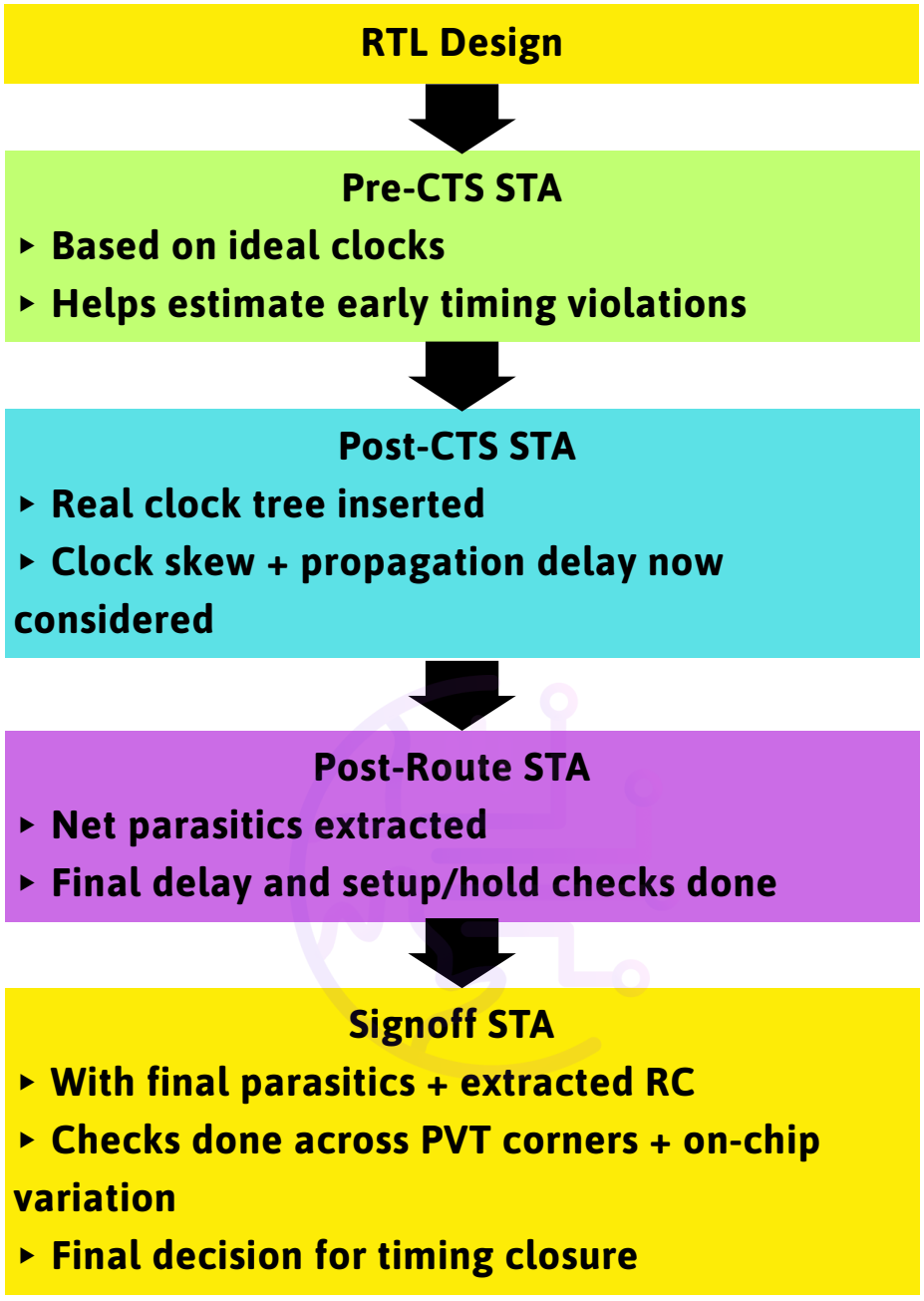
- **Includes cell delays (e.g., combinational logic) + net delays (wire delays).**

3. Capture Clock Edge

- **The clock edge at the destination flip-flop that captures the data.**

10. At what stages of the design flow is STA performed?

Company: Intel, Cadence



11. You see a negative setup slack in STA. How do you debug it?

Company: Synopsys, Qualcomm

Debug Checklist:

1.Path Type: Is it a real, false, or multicycle path?

→ False/multicycle paths can be ignored or relaxed.

2. High Cell Delay?

→ Replace slow gates or logic cones with faster cells (cell resizing).

3. Long Net Delay?

→ Check for long wires, routing congestion → fix with buffer insertion.

4. Clock Skew?

→ If capture clock arrives earlier than launch clock, try clock tree balancing.

5. Wrong Constraints?

→ Check if the path is constrained correctly using SDC (setup, hold, etc.).

12. What is the purpose of false paths and multicycle paths in STA?

Company: Apple, Samsung

False Path	Multicycle Path
Path that does not require timing check.	Path that needs more than 1 clock cycle to complete data transfer.
Used for test/control logic paths that don't switch every cycle.	Common in slow control logic or handshake paths.
Tells STA to ignore this path.	Set using set_multicycle_path command.

13. What's the meaning of 'Slack' in STA? How is it calculated?

Company: AMD, TSMC

Slack = Required Time – Arrival Time

Setup Slack (Worst-case delay):

- **Ensures data arrives before the capture clock edge.**

Formula:

Setup Slack = (T_clock + Setup Time) – (Data Delay + Launch Clock Delay)

Hold Slack (Best-case delay):

Ensures data doesn't arrive too early, causing race conditions.

Formula:

Hold Slack = (Data Delay + Launch Clock Delay) – (Hold Time + Capture Clock Delay)

Positive Slack = Good (no violation)

Negative Slack = Violation (needs fix)

14. Compare Setup vs Hold Timing Checks in STA

Company: Intel, MediaTek

Aspect	Setup Timing	Hold Timing
Check Type	Maximum delay check	Minimum delay check
When Checked	Ensures data arrives before capture clock	Ensures data doesn't arrive too early
Slack Formula	Required - Arrival (max)	Arrival (min) - Required
Violation Cause	Slow logic, long data path	Fast logic, short net delay, skew issues
Fixing Technique	Resize cells, buffer insertion, pipelining	Increase clock path delay
Critical for	High frequency designs	Low skew and functional

15. What is Crosstalk in STA and how is it analyzed?

Company: Apple, Qualcomm

- **Crosstalk is unwanted interference caused by switching activity on nearby nets (aggressors) impacting a victim net.**
- **It can introduce extra delay (crosstalk delay) or unintended transitions (crosstalk glitch).**
- **Analyzed using: Noise-aware STA tools like Primetime-SI, Tempus, which simulate coupling capacitance impact.**
- **Two key effects:**
  - **Crosstalk delay → changes timing paths (adds to setup/hold analysis).**
  - **Crosstalk glitch → may cause functional failures if glitches are captured.**

16. What are Clock Gating Checks in STA? Why are they important?

Company: AMD, Broadcom

Clock gating is used to save power by turning off clock to parts of the circuit when not needed.

In STA, tools perform clock gating checks to ensure:

- **No glitches pass through gating logic.**
- **Enable signal transitions are timed properly.**

It include:

- **Setup/Hold checks on enable paths.**
- **Glitch detection at AND/OR gates used for gating.**

Improper clock gating can cause:

- **Glitchy clocks → leading to false triggering of flops.**
- **Hold/setup violations due to enable signal misalignment.**

17. What is the role of Liberty (.lib) files in STA?

Company: ARM, Synopsys

- **Liberty files (.lib) define standard cell timing, power, and functionality.**
- **They are used by STA tools to calculate the delay and transition of signals in the netlist.**
- **Every standard cell has defined timing arcs (like cell_rise, cell_fall, rise_transition, etc.) for various input/output conditions.**
- **Delay is a function of:**
  - **Input transition (slew rate)**
  - **Output load (fanout or capacitance)**
- **These files also include:**
  - **Setup/Hold timing**
  - **Constraints (max transition, max capacitance)**
  - **Power models**
- **Tools like Primetime and Tempus use .lib to simulate delays under different PVT corners.**
- **Having accurate .lib is critical for reliable STA across multiple scenarios.**

18. Explain Timing Arcs and Types in STA

Company: Samsung, Microchip

- **Timing arcs represent how signals propagate through cells — from input pins to output pins.**
- **Each arc defines a delay (or constraint) between an input and output or between two pins.**
- **Two major types:**
- **Data Arcs: Describe combinational delays (e.g., NAND2: A → Y, B → Y).**
- **Constraint Arcs: Represent setup, hold, recovery, removal requirements (e.g., D to Q with respect to CLK).**

- **Timing arcs are library-defined, and used to build the timing graph of the design.**
- **STA tools use these arcs to perform path-based delay calculation.**
- **For sequential elements (flops, latches), constraint arcs play a vital role in launch-capture analysis.**
- **Arcs can be conditional (based on functionality) and are non-linear, requiring interpolation across .lib tables.**

19. What is Noise Modeling in STA and why is it important?

Company: Intel, Cadence

- **In modern STA, Noise Modeling refers to the simulation of unwanted signal activity (like crosstalk or switching noise) affecting timing accuracy.**
- **This is especially important in deep submicron nodes where coupling capacitance and aggressor-victim effects are significant.**
- **Noise can:**
- **Alter signal delay (making paths slower or faster).**
- **Introduce false transitions (glitches).**
- **Cause functional errors if glitches are captured by flip-flops.**

20. What is the difference between NLDM and CCS delay models?

Company: MediaTek, Renesas

NLDM (Non-Linear Delay Model):

- **Traditional model used in .lib files.**
- **Delay and transition are modeled using 2D lookup tables based on input slew and output load.**
- **Easy to implement, faster in computation.**
- **Not accurate in advanced nodes due to non-linear effects, especially with coupling.**

CCS (Composite Current Source) Model:

- **Advanced model that captures waveform shape and current flow.**
- **Provides more accurate delay and noise analysis, especially for crosstalk and multi-input switching.**
- **Requires detailed waveform-aware simulation, leading to higher runtime but better correlation with SPICE.**

CCS is now standard for signoff at 7nm and below, while NLDM is still used for preliminary timing closure.

21. What is Multi-Voltage Domain Timing Analysis in STA?

Company: Synopsys, Qualcomm

Multi-voltage design uses different voltage levels across blocks to reduce power.

In STA, this introduces complexity due to:

- **Timing violations when signals travel between different voltage domains.**
- **Delay variation caused by voltage differences.**

STA tools must ensure:

- **Correct timing across voltage domains.**
- **Level shifters are inserted and functional.**
- **Power intent files (UPF/CPF) are correctly interpreted.**

22. What are Isolation Cells and How Are They Verified in STA?

Company: Intel, Marvell

- **Isolation cells protect powered-off domains from corrupting the powered-on logic.**
- **They are inserted at domain boundaries and control what logic is passed during power-down states.**
- **These cells usually output a fixed logic value (0 or 1) when the domain is OFF.**

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23. What are the common causes of setup and hold violations in STA?

Company: Synopsys, AMD

- **Setup violation occurs when data arrives after the setup window closes.**
- **Hold violation happens when data arrives too early, before the hold window.**
- **Common causes include:**
  - **Unbalanced clock skew**
  - **Long combinational logic paths**
  - **High fanout or poor placement**
  - **Incorrect constraints or missing false/multicycle path definitions**
  - **Improper buffer sizing or over-optimization during synthesis**
- **Fixing Setup: Optimize logic, upsize cells, buffer nets, reduce path delay.**
- **Fixing Hold: Insert buffers to delay data, adjust clock tree, or modify constraints**

24. What is Negative Slack? How do you fix it?

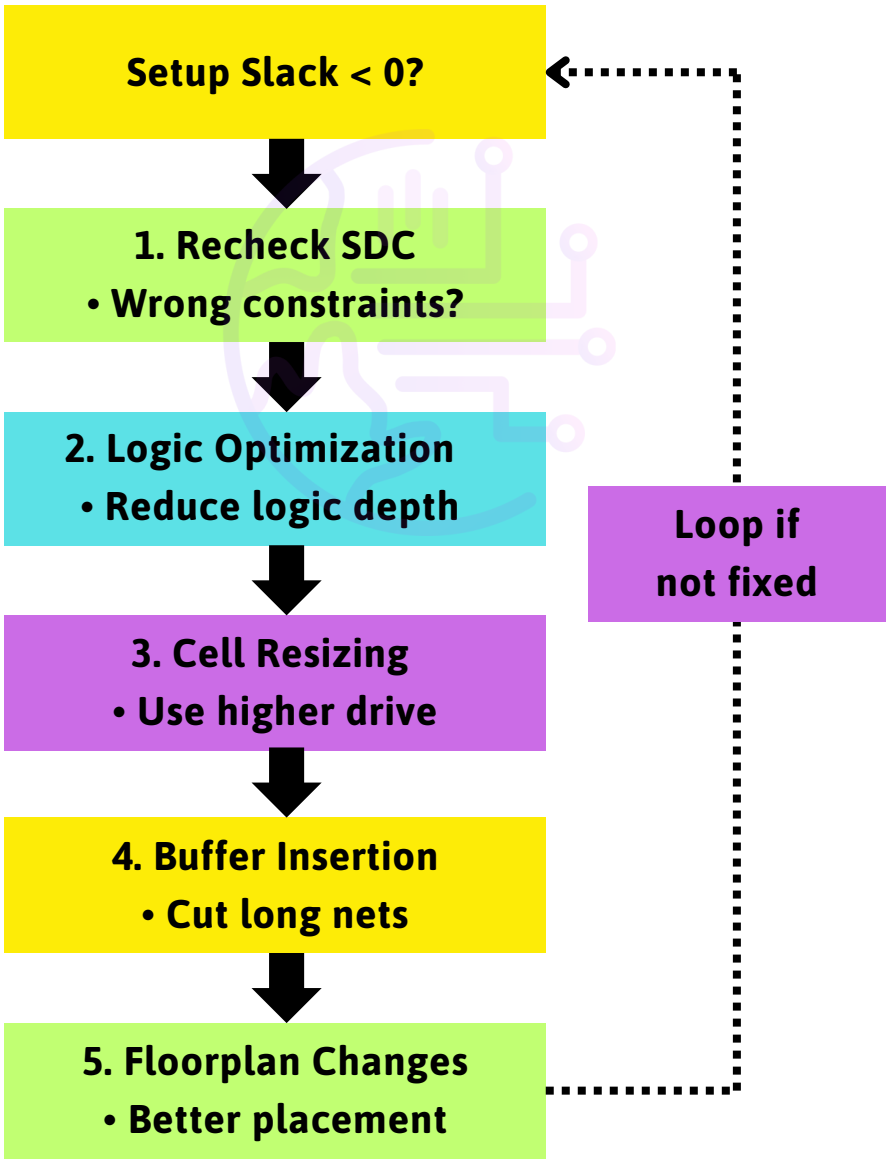
Company: Intel, Marvell

- **Negative Slack means the timing path is violating its required time.**
- **Setup Slack = Required Time – Arrival Time → if < 0 → Setup violation.**
- **Hold Slack = Arrival Time – Required Time → if < 0 → Hold violation.**

- To fix negative setup slack:
  - Reduce logic depth
  - Use faster cells
  - Optimize net delays
- To fix negative hold slack:
  - Add buffers or delay cells
  - Tweak clock tree to delay data

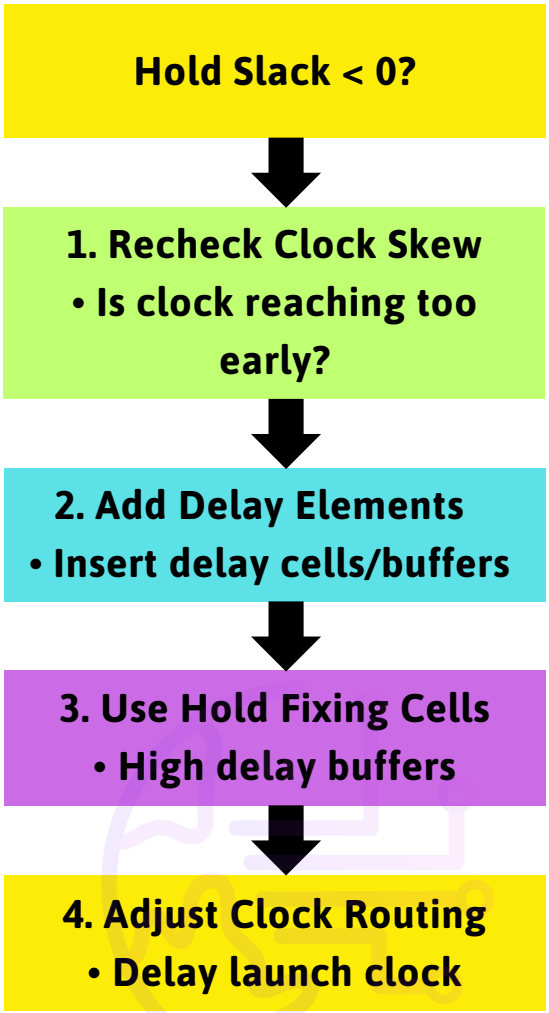
25. What is the flow to fix a Setup Violation?

Company: Synopsys, Samsung



26. What is the flow to fix a Hold Violation?

Company: Apple, NVIDIA



27. How does Process-Voltage-Temperature (PVT) variation impact STA signoff?

Company: Apple, Texas Instruments

- **PVT corners represent variations in process (fast/slow), voltage levels, and temperature across silicon.**
- **STA is done across multiple PVT corners to ensure timing holds in best-case and worst-case environments.**
- **Effects:**
  - **Process: Fast corner shortens delays, may cause hold violations; slow corner stretches delays, may cause setup violations.**
  - **Voltage: Lower voltage increases delay; higher voltage reduces delay but may cause hold issues.**

- Voltage: Lower voltage increases delay; higher voltage reduces delay but may cause hold issues.
- Temperature: Higher temperature → slower transistors, worse setup timing.

- Common corners: SS, FF, TT, SF, FS, each with specified voltage and temperature.
- Timing libraries (.lib) are provided for each PVT corner.
- STA tools analyze each corner independently and report violations.
- Signoff is accepted only when timing is clean across all PVT scenarios.

28. What is Clock Uncertainty in STA and how is it modeled?

Company: MediaTek, Marvell

Clock Uncertainty accounts for inaccuracies in clock edge arrival due to multiple factors.

Components:

- Jitter – Random variation due to PLLs, oscillators
- Skew – Design-related delay mismatch between launch and capture clocks
- Margin – Safety buffer added by designers

- STA models uncertainty by reducing the effective clock period available for data to propagate.
- Setup Uncertainty is subtracted from the launch-to-capture window.
- Hold Uncertainty is added to the hold requirement, making the timing window tighter.
- Clock uncertainty can vary across corners, modes, and domains.
- Over-conservative uncertainty can reduce performance, while underestimation leads to timing failures.

29. How are Integrated Clock Gating (ICG) cells handled in Static Timing Analysis?

Company: ARM, Synopsys

- **ICG cells are special clock gating elements used to reduce dynamic power by disabling the clock to inactive logic blocks.**
- **STA must verify both functional and timing correctness across enable transitions.**
- **Constraints used:**
  - **set_clock_gating_check -setup 0.1 -hold 0.05 [get_clocks clk]**
- **Synthesis tools insert ICG cells with enable logic; STA must trace the gating enable logic cone to analyze properly.**
- **Clock gating checks are included in timing reports and signoff flows.**
- **Misconfigured ICG paths can result in false negatives or positives in STA reports.**

30. What is the significance of Input/Output (IO) constraints in STA for top-level timing closure?

Company: Broadcom, Infineon

- **IO constraints define the timing relationship between internal clocks and external interfaces, critical for chip-level STA.**
- **Types of IO constraints:**
  - **set_input_delay – defines max delay allowed from external driver to input pin**
  - **set_output_delay – defines required delay from internal flop to IO pin**
- **These are specified relative to an external clock coming from the testbench or another die.**
- **Example constraints:**
  - **set_input_delay 1.2 -clock clk [get_ports in_data]**
  - **set_output_delay 1.0 -clock clk [get_ports out_data]**

31. How are Timing Exceptions like False Paths and Multicycle Paths handled in STA?

Company: Cadence, Micron

- **Timing exceptions inform the STA tool to ignore or alter timing checks on specific paths.**
- **False Path (FP): A path that never functions in real operation and should be excluded from analysis.**
  - **Example: asynchronous control lines or debug/test signals**
- **Constraint:**
  - **set_false_path -from [get_clocks clkA] -to [get_clocks clkB]**
- **Multicycle Path (MCP): A path that logically takes more than one clock cycle to transfer data.**
- **Used for low-speed logic or slow state machines**
- **Constraint:**
  - **set_multicycle_path 2 -setup -from X -to Y**
  - **set_multicycle_path 1 -hold -from X -to Y**
- **These exceptions override default 1-cycle setup/hold checks.**
- **Improper use may hide real violations, so must be verified by RTL and functional context.**
- **Reviewed carefully during timing signoff to avoid silicon bugs.**

32. How is STA performed on Asynchronous Clock Domains, and what are the key challenges?

Company: Intel, STMicroelectronics

- **In designs with multiple unrelated clocks, asynchronous paths occur between flip-flops controlled by different clock sources.**
- **STA cannot check setup/hold between asynchronous domains reliably due to lack of phase relationship.**

- challenges:
  - Metastability – receiver flip-flop may latch invalid data
  - Unknown data transfer time, so setup/hold checks are meaningless
- Solutions:
  - Use of synchronizers (2-stage flops) for single-bit control signals
  - Use of FIFO with gray code pointers for multi-bit data

33. Compare different types of STA corners used during timing closure

Company: Intel, STMicroelectronics

Type of Corner	Description	Impact on STA	Common Use Case
SS (Slow-Slow)	Slow process, low V, high T	Max delay → Setup issues	Worst-case setup check
FF (Fast-Fast)	Fast process, high V, low T	Min delay → Hold issues	Worst-case hold check
TT (Typical-Typical)	Typical process, voltage, temp	Nominal delays	Functional validation
SF / FS	Mixed fast/slow P corners	Path-specific timing	Extra margin paths
Low-Voltage	Lower voltage than nominal	Longer delays	Low-power analysis
High-Temperature	125°C or more	Slower transistors	Automotive/Military chips

34. What is CRPR (Clock Reconvergence Pessimism Removal) in STA and why is it important?

Company: AMD, Synopsys

- **CRPR removes unnecessary pessimism in timing paths that share the same clock network.**
- **When clock launch and capture paths reconverge, STA tools may double-count delay, making slack look worse.**
- **CRPR adjusts for shared clock path delays, improving slack accuracy.**
- **Without CRPR, tools assume independent clock delays, adding pessimism.**
- **Example: Two flops clocked by the same clock but different path segments — delay difference overestimated.**
- **Helps reduce false violations, especially in deep pipelines or large SoCs.**

35. How does On-Chip Variation (OCV) affect STA, and what is the difference between AOCV and POCV?

Company: NVIDIA, Samsung

- **OCV models timing variation across chip regions due to manufacturing/process differences.**
- **AOCV (Advanced OCV):**
- **Adds variation based on cell location, path depth, and library-specific tables**
- **More accurate than classic OCV**
- **POCV (Parametric OCV):**
- **Uses statistical data (sigma-based modeling)**
- **Captures more realistic variation across timing arcs**
- **Allows path-based analysis with tighter margins**
- **Both reduce over-pessimism, helping meet timing with better QoR.**

Tools support commands like:

- set_ocv_group -name groupA ...
- set_aocvm_table -delay ...
- set_pocvm_model ...

36. How is Static Timing Analysis handled in hierarchical or block-based design flows?

Company: Texas Instruments, Broadcom

- Large SoCs are divided into blocks/IPs, each analyzed separately.
- STA is done at block-level first, then at top-level (full chip).
- At block-level:
  - Use abstract views (black-box or timing models) of other blocks
  - Apply interface constraints (IO delays, virtual clocks)
- Top-level STA:
  - All blocks combined with top-level routing and real clocks
  - Re-validate full paths, including block-to-block timing
- Enables parallel development of blocks → faster convergence

37. What is Timing Budgeting in STA and how is it used during floorplanning?

Company: ARM, MediaTek

- Timing budgeting splits total available slack among sub-blocks or stages.
- Ensures each block has a target delay to meet full-chip timing
- Done early in floorplanning/placement to guide logic distribution
- Example:
- Total path delay target = 1ns
- Stage 1 budget = 0.4ns, Stage 2 = 0.3ns, Stage 3 = 0.3ns
- Helps avoid timing bottlenecks in a single stage
- Tools support auto budgeting and manual override based on criticality

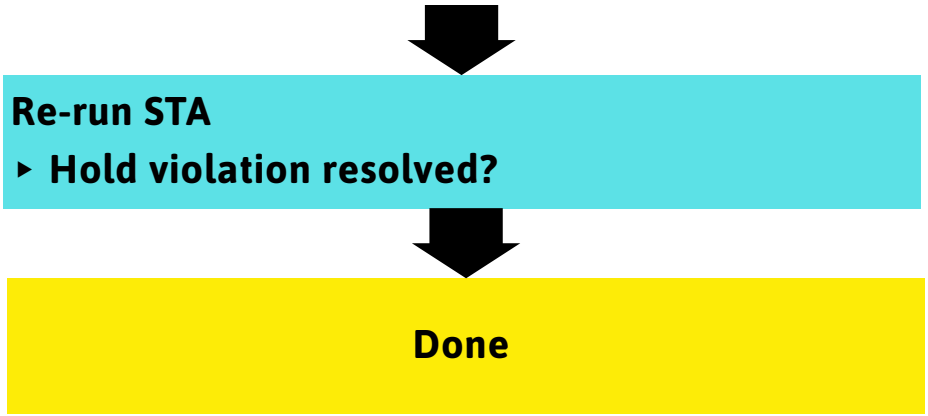
Command used (in some tools):

- **set_max_delay -from A -to B 0.4**

38. A path is showing a Hold Violation — How do you debug it?

Company: Intel, Synopsys





39. What is Crosstalk Noise in STA and how is it analyzed?

Company: Cadence, Micron

Crosstalk is an unwanted coupling between nearby signal nets (aggressor → victim).

- **In STA, it causes:**
- **Noise-induced delay → impacts setup timing.**
- **False transitions on victim nets → can lead to glitches.**
- **Tools calculate crosstalk delay and noise window based on:**
- **Coupling capacitance**
- **Switching direction (same or opposite)**
- **Signal slew and net spacing**
- **Crosstalk-aware STA includes noise analysis for both:**
- **Functional correctness (noise spikes),**
- **Timing closure (delay shifts).**
- **Fixes include shielding nets, spacing critical signals, or buffer insertion.**

40. What is Crosstalk Noise in STA and how is it analyzed?

Company: Intel, MediaTek

- **Clock Gating Cells (like ICGs) save power by turning off clocks to idle modules.**
- **They add extra timing arcs that must be validated during STA.**

- Tools perform checks between:
  - Enable signal to gate control logic
  - Gate control to clock pin
- Timing arcs from clock enable to output are crucial to avoid glitches.
- Setup/Hold analysis ensures:
  - Enable signal is stable before the active clock edge.
  - No glitch propagates to the gated clock.
- Constraints are added for gating checks, and synthesis tools usually infer these paths.

41. What are PVT Corners in STA and why are multiple corners used for signoff?

Company: Marvell, GlobalFoundries

- PVT = Process, Voltage, Temperature — variations that affect transistor behavior.
- Different PVT corners represent worst/best-case conditions:
  - Slow-Slow (SS): Worst-case speed (low temp/voltage) → used for setup analysis
  - Fast-Fast (FF): Fastest transistors (high temp/voltage) → used for hold analysis
  - Typical-Typical (TT): Nominal case for general estimation
- STA tools run timing checks across multiple PVT corners to ensure reliability.
- Example: A design must pass setup in SS@0.9V@125°C and hold in FF@1.1V@-40°C.
- Using multiple corners ensures:
  - Functional robustness across manufacturing variations.
  - Signoff readiness for any process condition.

42. How do you debug a Hold Violation in a timing report?

Company: Broadcom, Qualcomm

Hold violation → data is arriving too early at the capture flip-flop.

Common causes:

- **Short logic path**
- **Minimal data delay**
- **Skewed clock favoring early capture**

Debugging Steps:

- **Run report_timing -delay_type min in STA tool.**
- **Look for clock path delay vs. data path delay.**
- **Check if clock skew is increasing the violation.**
- **If net is too short → insert buffers or de-optimize fast gates.**

Final step: Re-run STA and verify Hold Slack ≥ 0 ps.

43. How do you debug a Hold Violation in a timing report?

Company: Renesas, STMicroelectronics

- **IO timing defines how signals interact with external components at chip boundaries.**
- **Important for top-level STA where design connects with board-level logic.**
- **Input constraints use set_input_delay, output with set_output_delay — both referenced to IO clock.**
- **IO timing affects:**
  - **Data setup/hold to/from external devices**
  - **Launch/capture alignment with interface clocks**
- **Signoff STA must:**
  - **Apply realistic board delays (trace delays, capacitance)**
  - **Account for IO buffer delays and pad timing**
- **Common mistake: Ignoring board skew and delay, leading to failed signoff.**

44. What is MMMC (Multi-Mode Multi-Corner) Analysis and why is it essential for timing signoff?

Company: Samsung, Synopsys

- **MMMC is the backbone of STA signoff — validates design across:**
  - **Multiple functional modes (e.g., functional, test, scan, low-power)**
  - **Multiple PVT corners (process, voltage, temperature)**
- **Each mode-corner combo has a unique set of constraints, timing libraries, and clocks.**
- **Designers create analysis views for each case:**
  - **Example:**
    - **view: func_SS_0.9V_125C**
    - **view: scan_TT_1.0V_25C**
- **STA is performed simultaneously across all views to detect worst-case delays.**
- **Helps avoid over-design, ensures robust performance, and speeds up ECO closure.**

MMMC = Mode × Corner × Library

45. How is Clock Jitter modeled in STA and what are its types?

Company: Nvidia, Apple

Clock Jitter = Variability in clock edge arrival time

- **It affects timing margins and must be modeled carefully in STA.**
- **Types of Jitter:**
  - **Cycle-to-Cycle Jitter: Variation between two consecutive edges.**
  - **Period Jitter: Change in the duration of clock periods.**
  - **Phase Jitter: Short-term deviation from ideal phase.**
  - **Random Jitter: Noise-induced and unpredictable.**

Modeling in STA:

- **Modeled using set_clock_uncertainty for both setup and hold.**
- **Higher jitter → lower effective timing window.**
- **STA tools use:**
- **Gaussian models for random jitter**
- **Systematic skew margins for deterministic jitter**

46. What are Clock Skew Balancing Techniques in STA and why are they critical?

Company: Qualcomm, Intel

Clock Skew = Difference in arrival time of the clock signal at launching vs. capturing flip-flop.

Techniques to Balance Skew:

- **H-Tree / Balanced Clock Trees: Symmetric distribution to reduce skew.**
- **Clock Buffers Insertion: Strategically placed buffers to match delays.**
- **Useful Skew: Intentionally adjusting skew to fix setup/hold violations.**
- **Clock Gating Aware CTS: Ensures gated elements are synchronized.**
- **CRPR (Clock Reconvergence Pessimism Removal): Removes pessimism caused by common clock paths.**

Significance:

- **High skew leads to timing violations (esp. hold).**
- **Advanced SoCs use skew-aware place & route tools.**
- **Used in low-power and high-frequency chips.**

47. How are ECO Fixes applied during STA timing closure?

Company: GlobalFoundries, Cadence

ECO (Engineering Change Order): Late-stage fixes without full re-synthesis.

Common ECO Fix Methods:

- **Cell Upsizing/Downsizing: Alters drive strength to reduce delay.**
- **Buffer Insertion/Removal: Adjusts path delay or isolation.**

- **Net Re-routing: Reduces parasitic delay or congestion.**
- **Gate Swapping: Replace logic cells with faster or better-fitting versions.**
- **Path Splitting: Parallelizes logic to meet timing.**

STA Tools Involved:

- **Primetime ECO, IC Compiler, Tempus ECO, Innovus ECO**
- **Constraints: set_eco_mode, write_script, etc.**

ECOs help finalize designs post signoff check, minimizing re-spin and time-to-market delays.

48. Key SDC Constraints in STA

Company: Synopsys, Analog Devices

Command	Used For	Example
create_clock	Define clock	create_clock -period 10 [get_ports clk]
set_input_delay	Input arrival time	set_input_delay 2.5 -clock clk [get_ports A]
set_output_delay	Output required time	set_output_delay 1.8 -clock clk [get_ports Z]
set_false_path	Ignore unused path	set_false_path -from A -to B
set_multicycle_path	Add cycles for data transfer	set_multicycle_path 2 -setup -from A -to B
set_clock_uncertainty	Add margin for skew/jitter	set_clock_uncertainty 0.2 [get_clocks clk]

49. What are Clock Skew Balancing Techniques in STA and why are they critical?

Company: Synopsys, Apple

Static Timing Analysis (STA) can be approached using two methodologies: Path-Based and Graph-Based.

- **Graph-Based STA is the most commonly used technique. It constructs a timing graph of the circuit and propagates arrival times, required times, and slacks through the graph using Breadth-First Search or similar algorithms. It is efficient and ideal for full-chip analysis and floorplan evaluation.**
- **Path-Based STA, on the other hand, involves evaluating specific timing paths end-to-end. This approach computes exact slack values by performing timing analysis only on critical paths. It's slower but more accurate and used during final signoff or for debugging violations.**

Summary:

- **Use Graph-Based STA for speed and coverage.**
- **Use Path-Based STA for precision and final checks.**

50. What is the role of Derating Factors in STA?

Company: Apple, Qualcomm, AMD

- **Derating factors adjust delays to model pessimism from variations in process, voltage, and temperature (PVT).**
- **They are applied to cell delays and net delays during STA.**

Types of Derating:

- **Global Derating: Uniform scaling across the chip.**
- **Path-Based Derating: Different values for launch and capture paths.**
- **AOCV/POCV: Advanced models for variation-aware timing.**

Why it's used:

- **Prevents over-optimistic timing.**
- **Ensures reliable silicon performance.**

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